01 / PROJECT / EVALUATOR GUIDE

English and Telugu Text-to-Speech

Two independent acoustic models. Explicit prosody prediction. A linked, evidence-based demonstration from dataset selection to generated speech.

Abstract

This assignment adapts ming024/FastSpeech2 for Svarah English and IndicTTS Telugu. A phoneme encoder, learned speaker embedding, duration/pitch/energy variance adaptor, length regulator and mel decoder form each acoustic model. A pretrained universal HiFi-GAN produces waveforms. This report explains the language changes, data protocol, training curves, all ten required inputs and the saved prosody predictions. It distinguishes current source behavior, historical trained artifacts and uncompleted listening judgments.

Evaluation artifact	Submitted evidence
Separate acoustic models	English: step 8,000, speaker 14; Telugu: step 13,000, speaker 0.
Training record	60,000 logged steps each; selected checkpoints match the minimum logged total validation loss.
Outputs	10 PCM WAVs, 30 variance PNGs, 4 loss PNGs and 4 CSV histories.
Interactive evaluation	Companion HTML embeds all ten WAVs, hoverable plots, exact text, token tables and listening-note export.

Start here: links and navigation

Main Drive: checkpoints and experiment artifacts | Colab pipeline | Repository and detailed README

Download/open the interactive HTML companion | Primary assignment PDF | Report build instructions

Requirement / section	Pages
Architecture, reference reuse and language changes	2-3
Data, alignment, training settings and complete process	4
English / Telugu convergence and instability	5-6 / 7-8
Every required sample: speech measurements, duration, F0, energy	9-13 English / 14-18 Telugu
Comparison, failures, limitations and improvements	19
Dependencies, configurations, commands and access	20

Evidence status: the committed outputs are inspectable. Speech pronunciation, intelligibility and naturalness remain unverified by a completed listening review. Reserve waveform-hash coverage and omitted runtime helpers prevent a claim of fully reproduced training/inference. Those gaps are recorded explicitly on pages 19-20.

02 / 1 / IMPLEMENTATION

Correct system architecture

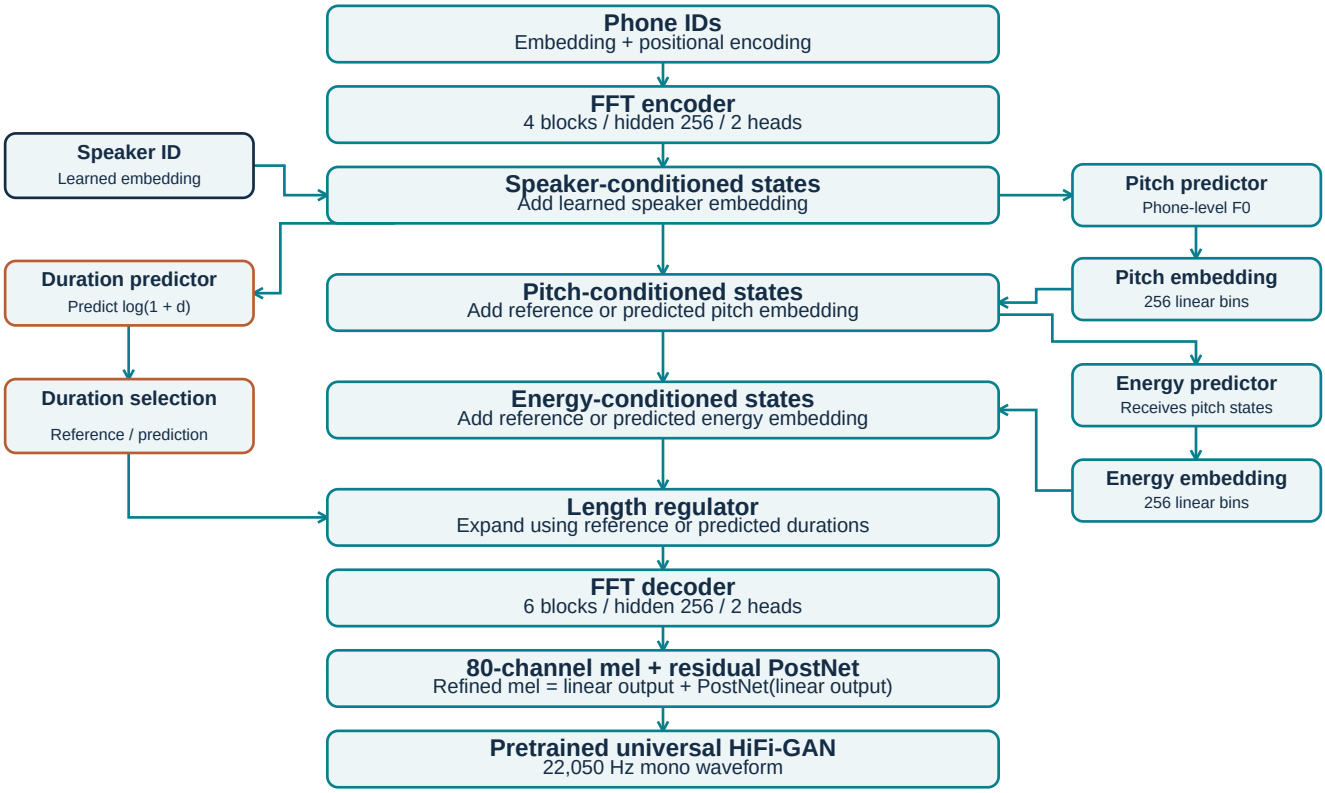


Figure 1. Each language uses this topology with independent acoustic weights and metadata. Duration and pitch receive speaker-conditioned states; energy receives pitch-conditioned states. Embeddings are added before length expansion. Arrows show actual model data flow.

Acoustic forward pass | Variance adaptor | Vocoder integration

Reference-conditioned training versus free-running inference

Training uses aligned durations and reference pitch/energy to condition mel reconstruction, while all three heads receive prediction losses. Inference supplies predicted duration, pitch and energy instead. The model is non-autoregressive after duration expansion. Low reference-conditioned mel loss therefore does not establish free-running intelligibility.

```
z_i target = log(1 + d_i)
d_hat_i = max(round(exp(z_i) - 1), 0)  # controls = 1
T_mel = sum_i(d_hat_i); seconds = T_mel * 256 / 22050
L = L1(mel, target) + L1(postnet_mel, target)
  + MSE(z, log(1+d)) + MSE(pitch, target) + MSE(energy, target)
```

Padding masks exclude padded phones/frames from loss. Pitch is interpolated PyWorld F0 averaged per phone and normalized. Energy is the phone average of the L2 norm of linear STFT magnitudes, not dB SPL. Plots reverse normalization using the matching statistics. Loss implementation

03 / 2 / IMPLEMENTATION

Reference reuse and language adaptations

Starting point: ming024/FastSpeech2. The report records reference d4e79eb; the original training commit and exact environment are not established. Current source fixes do not retroactively retrain the submitted checkpoint/audio artifacts.

Component	English adaptation	Telugu adaptation
Data preparation	Svarah loader; eight-field demographic profiles; filtering, disjoint pools and reserves.	IndicTTS loader; female/male source labels; independent filtering and reserves.
Frontend	Lexicon lookup plus g2p_en OOV fallback; number/time/date and abbreviation expansion.	Unicode phones: consonants, inherent vowel, vowel signs, virama, anusvara and visarga; Telugu numeric expansion.
Symbols / speakers	ARPAbet with lexical stress; learned profile embeddings. Profiles are not proven unique people.	Language-specific phone inventory; learned embeddings for two source voices.
Preprocessing	Interior silence/gaps retained; invalid spans/spn rejected; finite/length checks; training-only normalization and bounds.	Same checks with consistent Telugu lexicon/frontend boundaries; historical target mismatches remain.
Model / controls	FFT backbone retained; phone-level variance heads; energy-control routing and device-aware length regulator.	Same design, independent acoustic weights and statistics.
Training / inference	CSV/figure outputs, best-validation selection, explicit checkpoint/speaker paths and prediction JSON.	Independent language results with the same audit-friendly outputs.

Source map: English preparation / Telugu preparation; English cleaner / Telugu cleaner; alignment/features.

Differences from the supplied paper v8

Aspect	Paper v8	Project adaptation
Decoder / dropout	4 FFT blocks / 0.1	6 FFT blocks / 0.2
Pitch / energy	Frame CWT pitch / frame energy	Direct phone-level F0 / phone energy
Output / vocoder	No added PostNet / Parallel WaveGAN	Residual PostNet / universal HiFi-GAN
Batch / training steps	48 / 160k for paper FastSpeech 2	16 / 60k logged per language

Paper comparison: supplied arXiv v8, Section 2.3 and Appendix A-B. This is FastSpeech 2 acoustic synthesis with a separate vocoder, not the paper's direct-waveform FastSpeech 2s. English first-pronunciation lookup can miss contextual stress; Telugu decomposition is a phonemic approximation with allophone/loanword limits.

04 / 3 / DATA & EXPERIMENT

Data-to-speech process and training setup

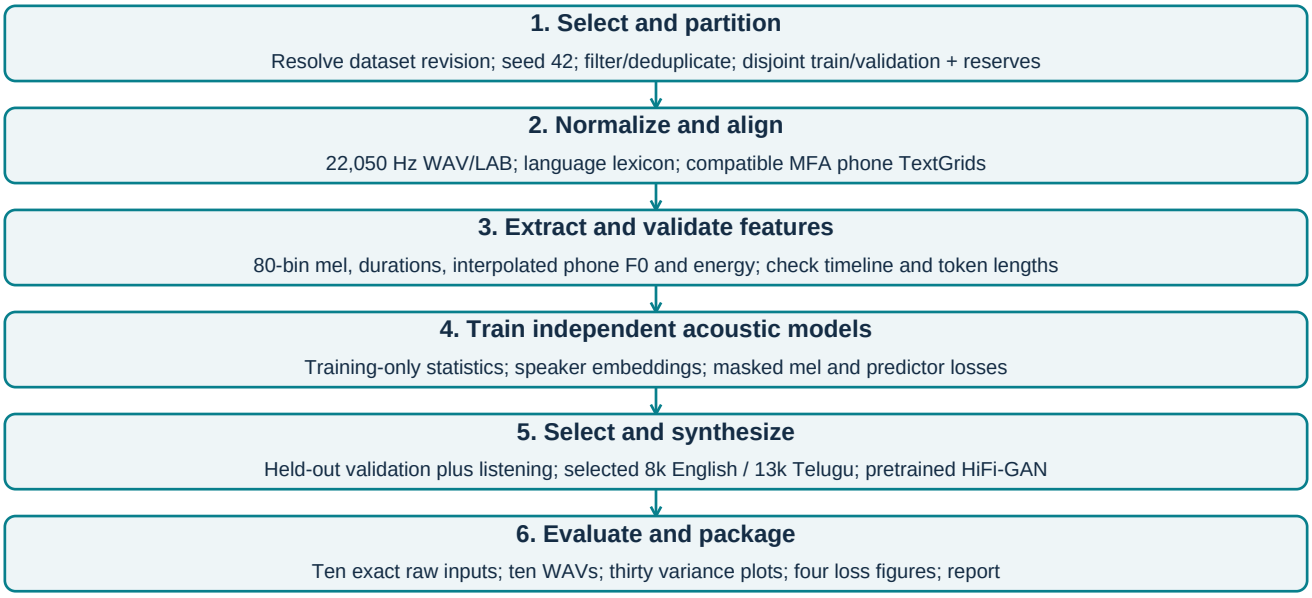


Figure 2. Complete intended lifecycle. The saved outputs are historical artifacts; reproduction requires the omitted helpers and matching bundles described on page 20.

Retained data (recovered evidence)	English / Svarah	Telugu / IndicTTS
Training utterances / retained hours	3,359 / 5.000 h	2,164 / 5.000 h
Validation utterances / hours	674 / 1.001 h	421 / 1.000 h
Training / validation groups	115 / 114 demographic profiles	2 / 2 source voices
ID overlap / missing final audio hashes	0 / 382 reserve utterances	0 / 339 reserve utterances

Selection resolves a dataset revision, uses seed 42, filters 1-20 s clips, writes 22,050 Hz mono PCM/LAB files and deduplicates waveform SHA-1 values. Voice/profile-stratified pools are shuffled and split about 5:1; separate 0.75h/0.20h reserves replenish rejected train/validation clips. Svarah's source pool is its test split; the assignment partitions are created from that pool. Recovered manifests confirm zero ID overlap and zero overlap among known hashes; missing reserve hashes prevent a complete final waveform-level claim.

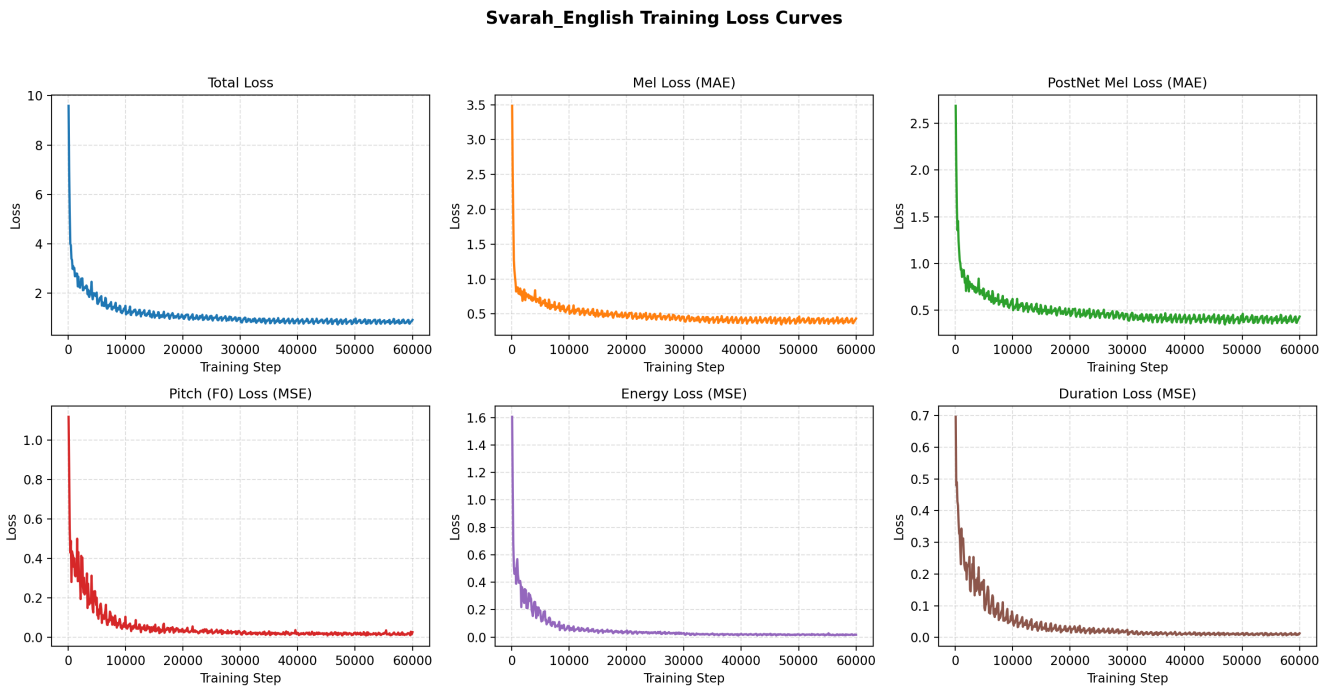
MFA phone TextGrids must precede feature extraction. Compatible lexicon/acoustic phones and unchanged waveform/transcript correspondence are required. Current code preserves interior pauses and checks the full aligned timeline before slicing. PyWorld F0 is interpolated and phone-averaged; energy is phone-averaged STFT magnitude. Selected training examples alone determine current normalization and quantization bounds.

Configuration: 4 encoder / 6 decoder FFT blocks; hidden 256; 2 heads; convolution filter 1,024, kernels 9/1; predictor channels 256, kernel 3, dropout 0.5; 256 linear pitch/energy bins; 80 mel channels, FFT/window 1,024, hop 256, 0-8 kHz. Adam betas (0.9, 0.98), epsilon 1e-9, batch 16, clipping 1.0, warmup 4k, anneal 30k/45k/55k by 0.3, total 60k. These are checked-in defaults, not a recovered environment lock.

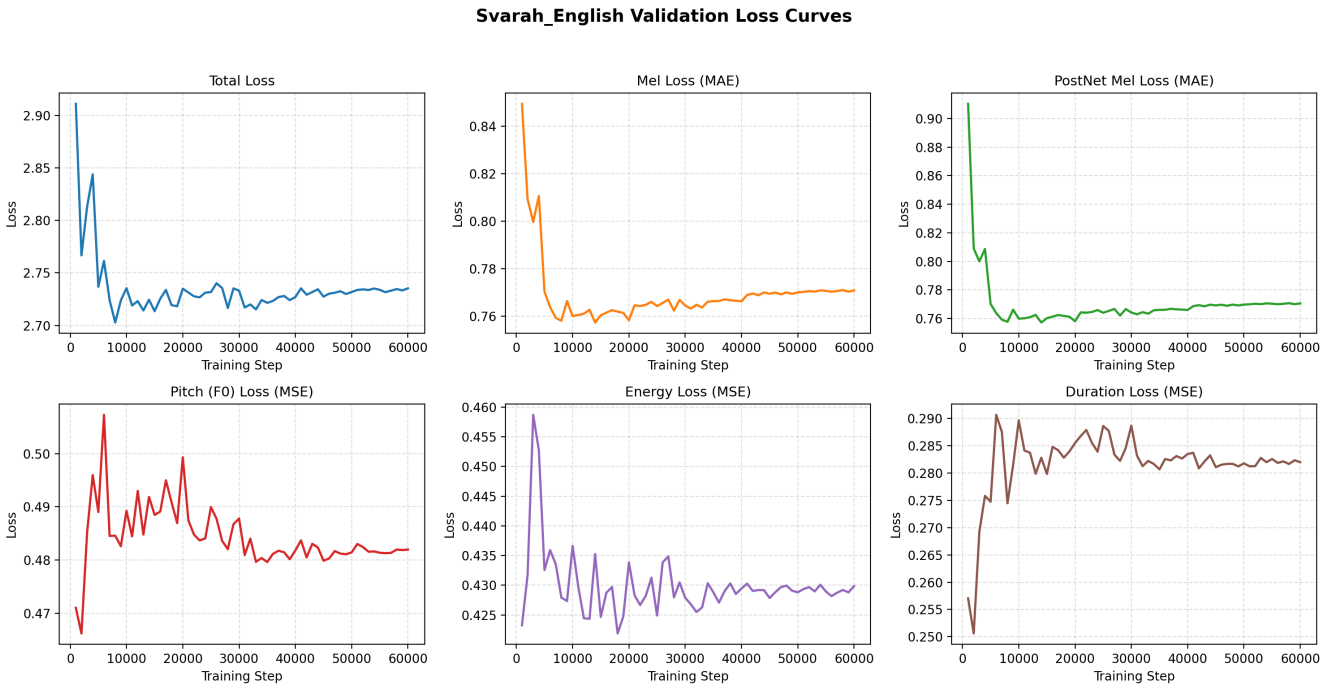
Dataset: Svarah | Dataset: IndicTTS Telugu | English config / Telugu config | Recovered-evidence summary

05 / 4 / CONVERGENCE

English: training and validation curves



English training: total, mel, PostNet, duration, pitch and energy losses. Original submitted figure, original aspect ratio.



English validation: total, mel, PostNet, duration, pitch and energy losses. Original submitted figure, original aspect ratio.

Training continues to decrease while validation improves mainly early and then plateaus. The selected validation minimum occurs before the final 60k step. No exploding or non-finite loss is present in the supplied CSVs; early fluctuations and the persistent generalization gap remain visible. Low losses alone cannot establish correct free-running speech.

Training CSV | Validation CSV | Numerical analysis

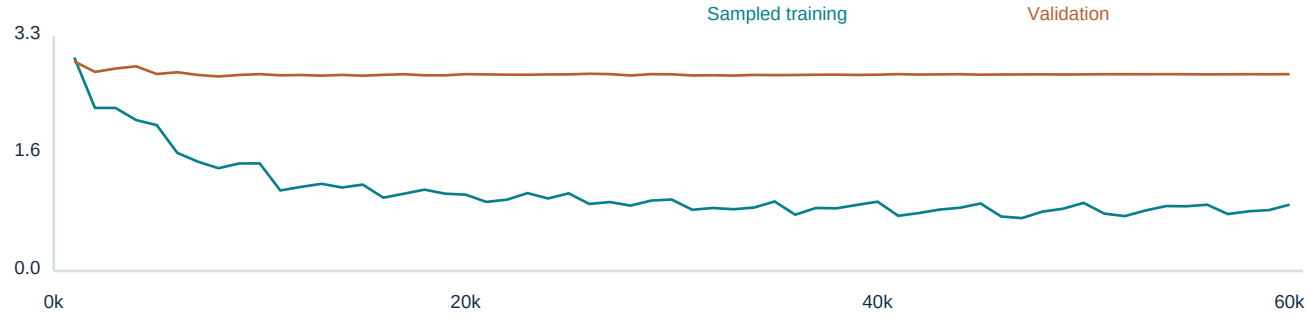
06 / 4 / NUMERICAL ANALYSIS

English: checkpoint selection and overfitting

Record / step	Total	Mel	PostNet	Pitch	Energy	Duration
Train first 100	9.5892	3.4841	2.6863	1.1178	1.6051	0.6959
Train selected 8,000	1.4274	0.5931	0.5930	0.0854	0.0776	0.0783
Train final 60,000	0.9216	0.4331	0.4326	0.0259	0.0168	0.0131
Val first 1,000	2.9111	0.8495	0.9103	0.4710	0.4233	0.2570
Val minimum 8,000	2.7028	0.7581	0.7577	0.4846	0.4279	0.2745
Val final 60,000	2.7353	0.7709	0.7706	0.4820	0.4299	0.2820

The histories contain 600 training rows and 60 validation rows. Minimum total validation loss is **2.702820 at step 8,000**: 7.15% below the first validation value. The final validation total is 2.735320, 1.20% above that minimum. The submitted prediction JSON identifies this selected step.

Generalization gap at validation checkpoints



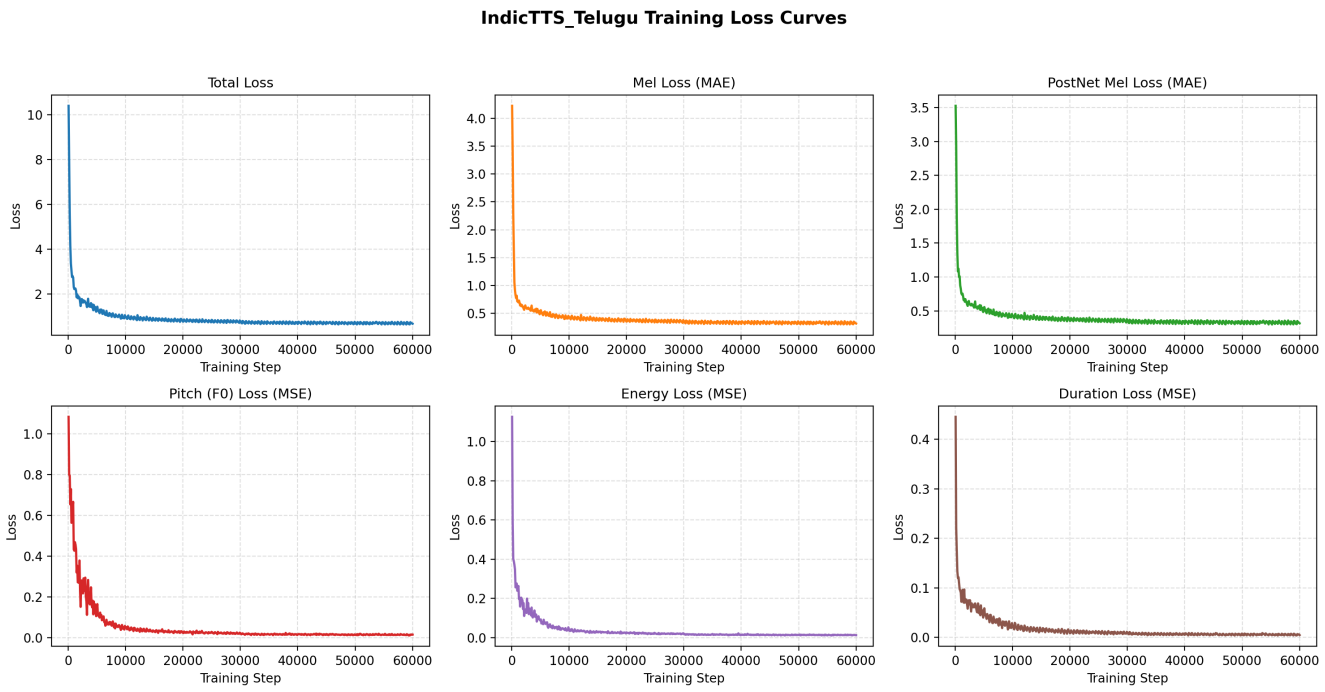
The continuing fall in training error alongside a nearly flat or rising validation curve is consistent with overfitting. Mel and PostNet validation losses no longer track the reduction in training losses. The early checkpoint is the best by the logged total-loss rule; perceptual checkpoint selection remains provisional until held-out free-running audio is reviewed.

Training rows are sampled minibatches, whereas validation is aggregated over held-out data. Their difference is descriptive, not a matched-sample confidence estimate. The component losses mix normalized features and mel error; absolute totals across English and Telugu cannot rank naturalness. Hardware, elapsed training time and an exact historical dependency lock are not established by these CSVs.

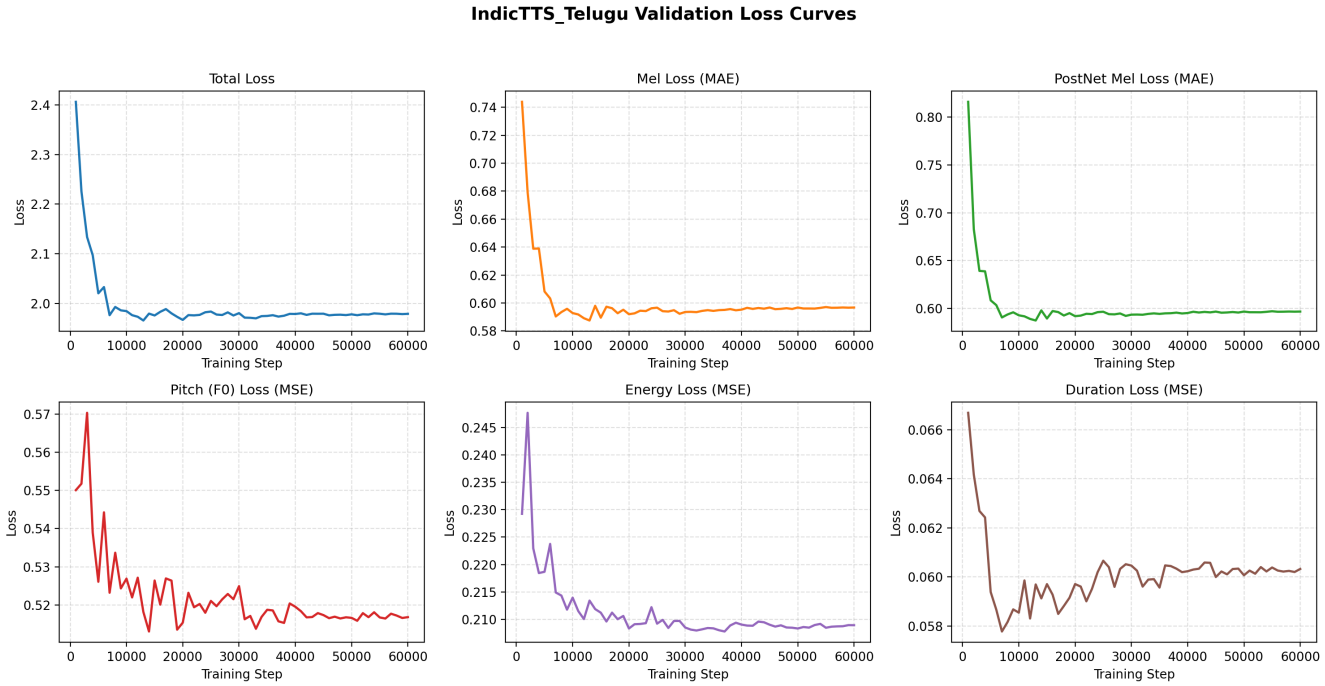
The total loss is the unweighted sum of the five displayed terms. Validation occurs every 1,000 steps in the supplied histories. No loss smoothing or invented intermediate measurements are used in these tables.

07 / 4 / CONVERGENCE

Telugu: training and validation curves



Telugu training: total, mel, PostNet, duration, pitch and energy losses. Original submitted figure, original aspect ratio.



Telugu validation: total, mel, PostNet, duration, pitch and energy losses. Original submitted figure, original aspect ratio.

Training continues to decrease while validation improves mainly early and then plateaus. The selected validation minimum occurs before the final 60k step. No exploding or non-finite loss is present in the supplied CSVs; early fluctuations and the persistent generalization gap remain visible. Low losses alone cannot establish correct free-running speech.

Training CSV | Validation CSV | Numerical analysis

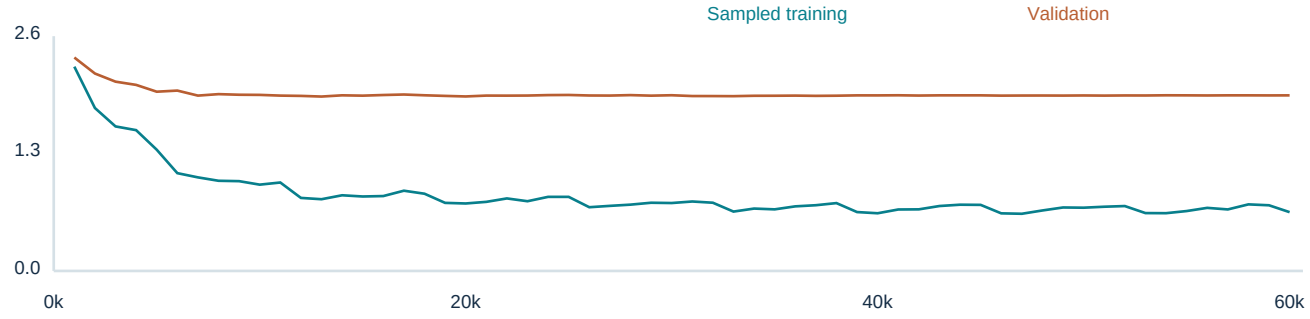
08 / 4 / NUMERICAL ANALYSIS

Telugu: checkpoint selection and overfitting

Record / step	Total	Mel	PostNet	Pitch	Energy	Duration
Train first 100	10.4081	4.2251	3.5288	1.0831	1.1258	0.4454
Train selected 13,000	0.8086	0.3687	0.3684	0.0321	0.0284	0.0111
Train final 60,000	0.6627	0.3153	0.3150	0.0152	0.0126	0.0045
Val first 1,000	2.4060	0.7439	0.8161	0.5500	0.2292	0.0667
Val minimum 13,000	1.9660	0.5875	0.5872	0.5182	0.2134	0.0597
Val final 60,000	1.9794	0.5968	0.5966	0.5168	0.2089	0.0603

The histories contain 600 training rows and 60 validation rows. Minimum total validation loss is **1.965954 at step 13,000**: 18.29% below the first validation value. The final validation total is 1.979367, 0.68% above that minimum. The submitted prediction JSON identifies this selected step.

Generalization gap at validation checkpoints



The continuing fall in training error alongside a nearly flat or rising validation curve is consistent with overfitting. Mel and PostNet validation losses no longer track the reduction in training losses. The early checkpoint is the best by the logged total-loss rule; perceptual checkpoint selection remains provisional until held-out free-running audio is reviewed.

Training rows are sampled minibatches, whereas validation is aggregated over held-out data. Their difference is descriptive, not a matched-sample confidence estimate. The component losses mix normalized features and mel error; absolute totals across English and Telugu cannot rank naturalness. Hardware, elapsed training time and an exact historical dependency lock are not established by these CSVs.

The total loss is the unweighted sum of the five displayed terms. Validation occurs every 1,000 steps in the supplied histories. No loss smoothing or invented intermediate measurements are used in these tables.

09 / 5 / PER-SENTENCE EVALUATION

E1: Rare long word and completeness

English | acoustic step 8,000 | speaker 14 | Play / download WAV | Raw text and arrays

Raw assignment input:

The antidisestablishmentarianism movement gained considerable momentum during the parliamentary proceedings.

Saved normalized model input:

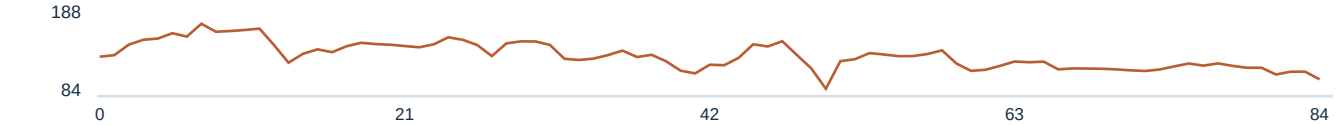
the antidisestablishmentarianism movement gained considerable momentum during the parliamentary proceedings

Measured waveform / prediction	Value
Tokens / frames / WAV duration	85 / 501 / 5.817 s
F0 mean [range]; energy mean [range]	134.65 [93.11, 174.44] Hz; 77.90 [13.95, 149.04]
Duration median / max; zero phones; pause frames	6 / 17; 1; 0

Duration / mel frames

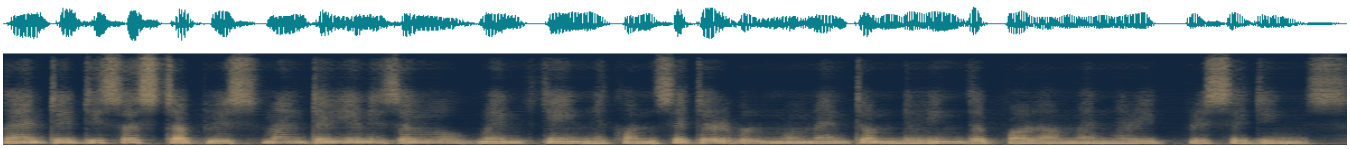


Predicted phone F0 / Hz



Predicted phone energy / STFT units





Waveform + 0-8 kHz spectrogram; Hann 1,024 / hop 128; colors are dB relative to utterance peak.

Measured audio: RMS -16.8 dBFS; peak -3.2 dBFS; 0 full-scale samples. Nonzero energy and no full-scale saturation do not establish intelligibility.

Sentence-specific analysis: The long word antidisestablishmentarianism stresses lexical coverage and articulation. Token 50 (AH0) has zero predicted frames and is therefore removed by length expansion. This is a concrete token-level omission risk; it does not establish that a whole audible word is missing.

Duration: median 6 frames; longest 17 frames (0.197 s) at token 84. **F0:** peak 174.44 Hz at token 7; first/last five-token means 145.4/112.8 Hz. **Energy:** maximum 149.04 at token 38, minimum 13.95 at token 76. These positions identify timing/emphasis candidates and weak-phone/pause regions for listening, rather than proving lexical stress. Token means are unweighted; phone identity and F0 interpolation affect the curves.

Pronunciation / overall quality: listening review pending. Inspect the target words, omissions/repetitions, naturalness and phrase timing in the exact linked WAV; use the interactive companion to record notes. Predicted F0 is phone-level and interpolated, not a measured generated-waveform F0 track.

Original labeled plots: Duration | Pitch | Energy | All sample links

10 / 5 / PER-SENTENCE EVALUATION

E2a: Noun-context conduct

English | acoustic step 8,000 | speaker 14 | Play / download WAV | Raw text and arrays

Raw assignment input:

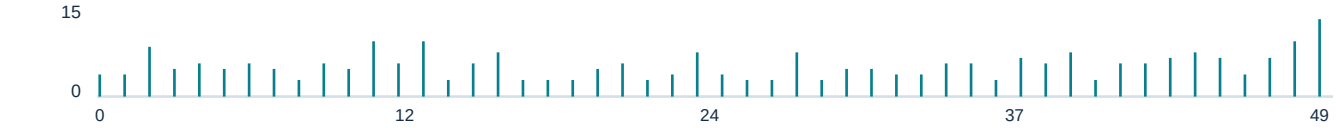
The orchestra's conduct during the ceremony was praised by the audience.

Saved normalized model input:

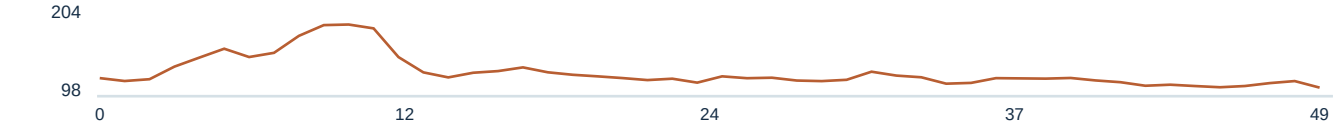
the orchestra's conduct during the ceremony was praised by the audience

Measured waveform / prediction	Value
Tokens / frames / WAV duration	50 / 283 / 3.286 s
F0 mean [range]; energy mean [range]	128.44 [109.06, 189.23] Hz; 68.40 [7.16, 138.10]
Duration median / max; zero phones; pause frames	5.5 / 14; 0; 0

Duration / mel frames

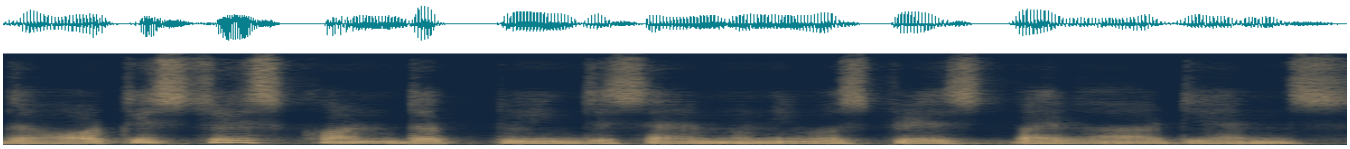


Predicted phone F0 / Hz



Predicted phone energy / STFT units





Waveform + 0-8 kHz spectrogram; Hann 1,024 / hop 128; colors are dB relative to utterance peak.

Measured audio: RMS -17.9 dBFS; peak -2.6 dBFS; 0 full-scale samples. Nonzero energy and no full-scale saturation do not establish intelligibility.

Sentence-specific analysis: The saved phone sequence contains the noun-stressed CONDUCT pronunciation K AA1 N D AH0 K T. Compare E2b: the same stress sequence is used there. The two contexts therefore do not receive distinct lexical stress labels from this saved frontend; final audible stress still needs listening.

Duration: median 5.5 frames; longest 14 frames (0.163 s) at token 49. **F0:** peak 189.23 Hz at token 10; first/last five-token means 128.3/112.4 Hz. **Energy:** maximum 138.10 at token 9, minimum 7.16 at token 17. These positions identify timing/emphasis candidates and weak-phone/pause regions for listening, rather than proving lexical stress. Token means are unweighted; phone identity and F0 interpolation affect the curves.

Pronunciation / overall quality: listening review pending. Inspect the target words, omissions/repetitions, naturalness and phrase timing in the exact linked WAV; use the interactive companion to record notes. Predicted F0 is phone-level and interpolated, not a measured generated-waveform F0 track.

Original labeled plots: Duration | Pitch | Energy | All sample links

11 / 5 / PER-SENTENCE EVALUATION

E2b: Verb-context conduct

English | acoustic step 8,000 | speaker 14 | Play / download WAV | Raw text and arrays

Raw assignment input:

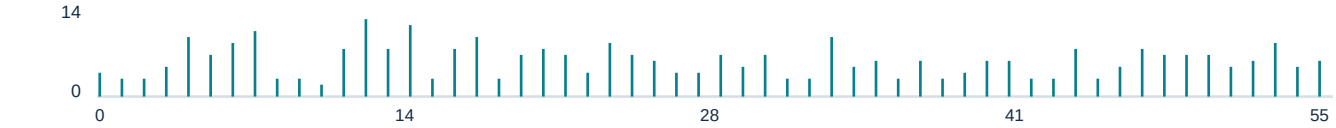
The researchers will conduct an experiment to evaluate the proposed method.

Saved normalized model input:

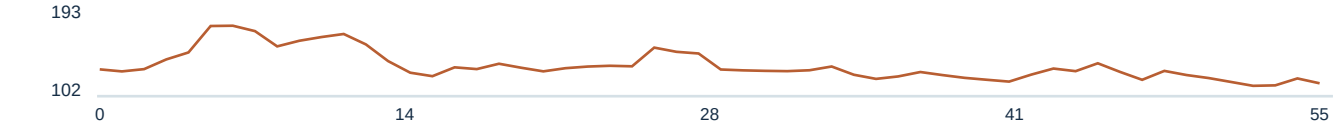
the researchers will conduct an experiment to evaluate the proposed method

Measured waveform / prediction	Value
Tokens / frames / WAV duration	56 / 337 / 3.913 s
F0 mean [range]; energy mean [range]	134.79 [112.90, 178.28] Hz; 67.33 [9.44, 121.10]
Duration median / max; zero phones; pause frames	6 / 13; 0; 0

Duration / mel frames

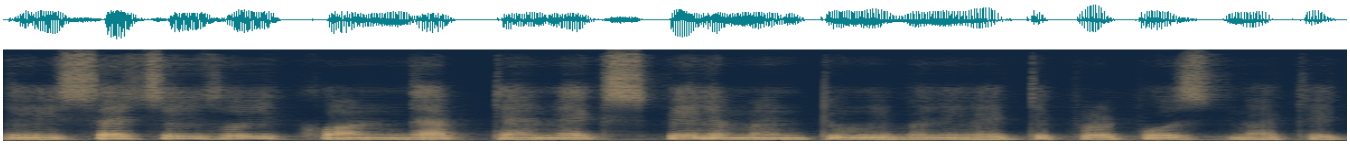


Predicted phone F0 / Hz



Predicted phone energy / STFT units





Waveform + 0-8 kHz spectrogram; Hann 1,024 / hop 128; colors are dB relative to utterance peak.

Measured audio: RMS -17.7 dBFS; peak -2.1 dBFS; 0 full-scale samples. Nonzero energy and no full-scale saturation do not establish intelligibility.

Sentence-specific analysis: The verb-context sentence uses the same K AA1 N D AH0 K T sequence as E2a, rather than the alternate dictionary entry K AH0 N D AH1 K T. This is an observed frontend stress limitation. Contextual acoustic differences cannot substitute for checking the intended pronunciation.

Duration: median 6 frames; longest 13 frames (0.151 s) at token 12. **F0:** peak 178.28 Hz at token 6; first/last five-token means 136.3/116.0 Hz. **Energy:** maximum 121.10 at token 34, minimum 9.44 at token 32. These positions identify timing/emphasis candidates and weak-phone/pause regions for listening, rather than proving lexical stress. Token means are unweighted; phone identity and F0 interpolation affect the curves.

Pronunciation / overall quality: listening review pending. Inspect the target words, omissions/repetitions, naturalness and phrase timing in the exact linked WAV; use the interactive companion to record notes. Predicted F0 is phone-level and interpolated, not a measured generated-waveform F0 track.

Original labeled plots: Duration | Pitch | Energy | All sample links

12 / 5 / PER-SENTENCE EVALUATION

E3: Broken word: rhyth m

English | acoustic step 8,000 | speaker 14 | Play / download WAV | Raw text and arrays

Raw assignment input:

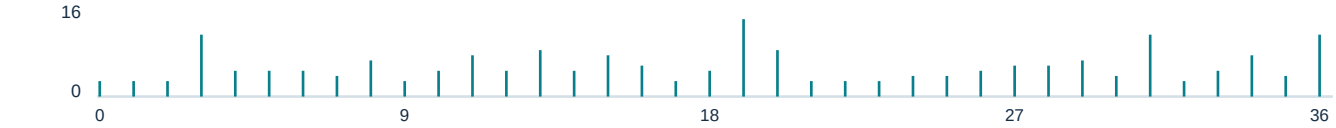
The rhyth m of the orchestra changed unexpectedly.

Saved normalized model input:

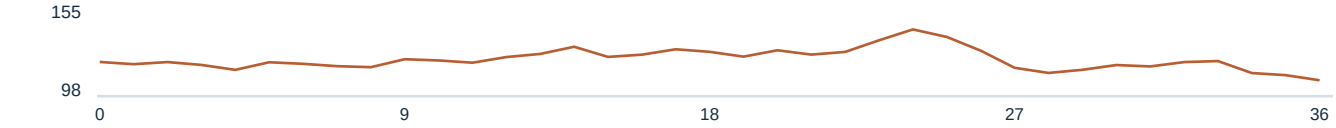
the rhyth m of the orchestra changed unexpectedly

Measured waveform / prediction	Value
Tokens / frames / WAV duration	37 / 217 / 2.519 s
F0 mean [range]; energy mean [range]	122.93 [108.71, 143.44] Hz; 52.73 [6.44, 113.02]
Duration median / max; zero phones; pause frames	5 / 15; 0; 0

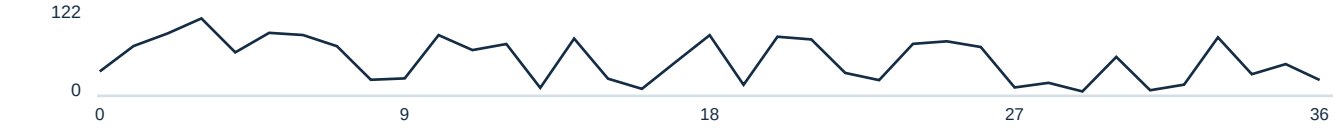
Duration / mel frames

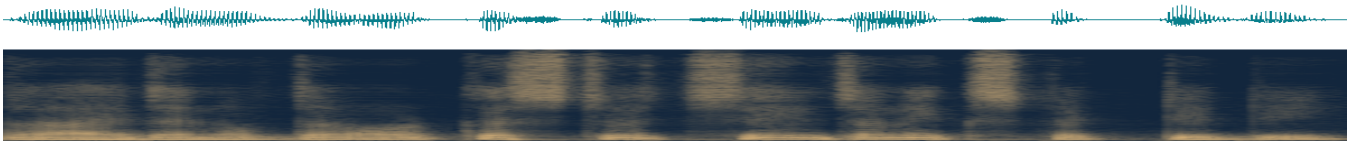


Predicted phone F0 / Hz



Predicted phone energy / STFT units





Waveform + 0-8 kHz spectrogram; Hann 1,024 / hop 128; colors are dB relative to utterance peak.

Measured audio: RMS -18.8 dBFS; peak -4.5 dBFS; 0 full-scale samples. Nonzero energy and no full-scale saturation do not establish intelligibility.

Sentence-specific analysis: The raw input retains the assignment double space inside rhyth m. Its normalized form is rhyth m: repeated whitespace collapses, but the word boundary remains. There are no explicit silence tokens. The relatively narrow F0 span suggests less predicted pitch variation; it does not prove monotone audible speech.

Duration: median 5 frames; longest 15 frames (0.174 s) at token 19. **F0:** peak 143.44 Hz at token 24; first/last five-token means 119.4/115.5 Hz. **Energy:** maximum 113.02 at token 3, minimum 6.44 at token 29. These positions identify timing/emphasis candidates and weak-phone/pause regions for listening, rather than proving lexical stress. Token means are unweighted; phone identity and F0 interpolation affect the curves.

Pronunciation / overall quality: listening review pending. Inspect the target words, omissions/repetitions, naturalness and phrase timing in the exact linked WAV; use the interactive companion to record notes. Predicted F0 is phone-level and interpolated, not a measured generated-waveform F0 track.

Original labeled plots: Duration | Pitch | Energy | All sample links

13 / 5 / PER-SENTENCE EVALUATION

E4: Time, date, loanword and pauses

English | acoustic step 8,000 | speaker 14 | Play / download WAV | Raw text and arrays

Raw assignment input:

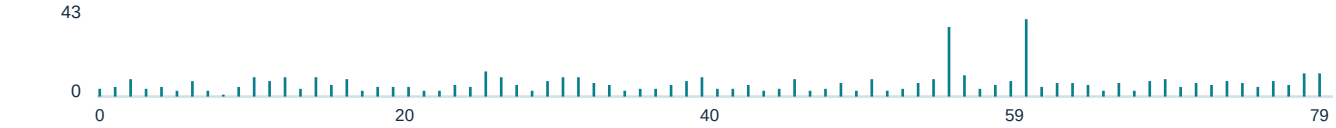
The meeting will start at 10:45 AM on 12 November 2026, so andaroo should arrive on time.

Saved normalized model input:

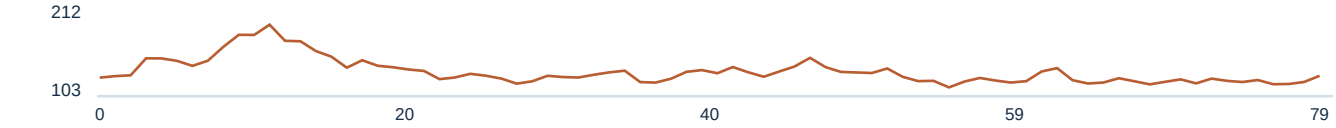
the meeting will start at ten forty five a m on twelve november twenty twenty-six, so andaroo should arrive on time

Measured waveform / prediction	Value
Tokens / frames / WAV duration	80 / 562 / 6.525 s
F0 mean [range]; energy mean [range]	135.15 [114.78, 196.37] Hz; 68.39 [5.23, 139.49]
Duration median / max; zero phones; pause frames	6 / 40; 0; 76

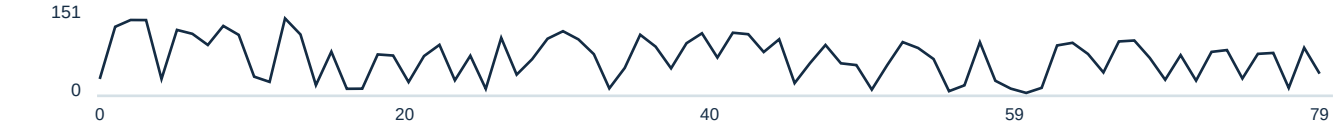
Duration / mel frames

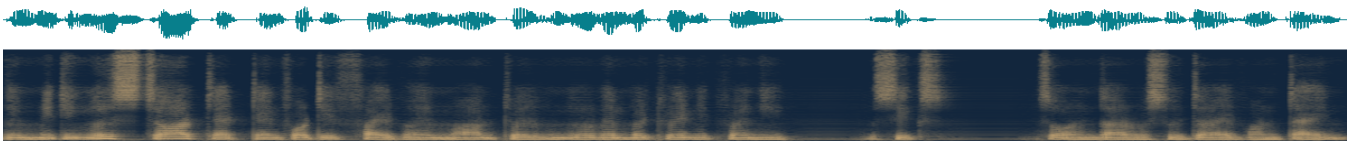


Predicted phone F0 / Hz



Predicted phone energy / STFT units





Waveform + 0-8 kHz spectrogram; Hann 1,024 / hop 128; colors are dB relative to utterance peak.

Measured audio: RMS -19.0 dBFS; peak -2.1 dBFS; 0 full-scale samples. Nonzero energy and no full-scale saturation do not establish intelligibility.

Sentence-specific analysis: The saved normalization expands 10:45 AM and the date, retaining andaroo. Two sp tokens occur at indices 55 and 60. Inspection of token neighbors places index 55 at the hyphen in twenty-six and index 60 at the comma: treating the numeric hyphen as punctuation can create an unintended pause inside the year.

Duration: median 6 frames; longest 40 frames (0.464 s) at token 60. **F0:** peak 196.37 Hz at token 11; first/last five-token means 138.5/122.8 Hz. **Energy:** maximum 139.49 at token 12, minimum 5.23 at token 60. These positions identify timing/emphasis candidates and weak-phone/pause regions for listening, rather than proving lexical stress. Token means are unweighted; phone identity and F0 interpolation affect the curves.

Pronunciation / overall quality: listening review pending. Inspect the target words, omissions/repetitions, naturalness and phrase timing in the exact linked WAV; use the interactive companion to record notes. Predicted F0 is phone-level and interpolated, not a measured generated-waveform F0 track.

Original labeled plots: Duration | Pitch | Energy | All sample links

14 / 5 / PER-SENTENCE EVALUATION

T1: Long technical word and clusters

Telugu | acoustic step 13,000 | speaker 0 | Play / download WAV | Raw text and arrays

Raw assignment input:

న్యూరోఫార్మకోలాజికల్ పరిశోధనలో నూతన ఔషధాల ప్రభావాన్ని విశ్లేషించారు.

Saved normalized model input:

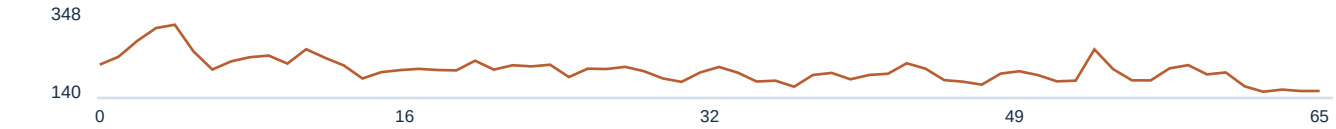
న్యూరోఫార్మకోలాజికల్ పరిశోధనలో నూతన ఔషధాల ప్రభావాన్ని విశ్లేషించారు.

Measured waveform / prediction	Value
Tokens / frames / WAV duration	66 / 451 / 5.236 s
F0 mean [range]; energy mean [range]	209.35 [155.40, 321.95] Hz; 58.79 [8.16, 140.61]
Duration median / max; zero phones; pause frames	6.5 / 16; 0; 0

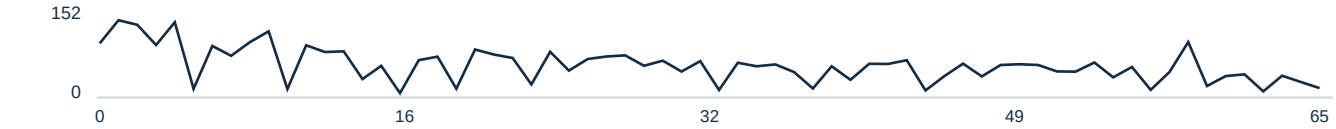
Duration / mel frames

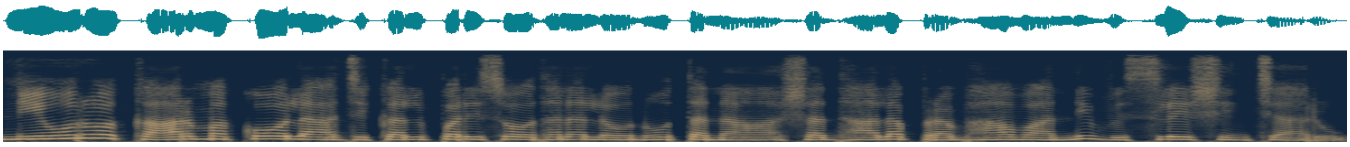


Predicted phone F0 / Hz



Predicted phone energy / STFT units





Waveform + 0-8 kHz spectrogram; Hann 1,024 / hop 128; colors are dB relative to utterance peak.

Measured audio: RMS -16.5 dBFS; peak -2.7 dBFS; 0 full-scale samples. Nonzero energy and no full-scale saturation do not establish intelligibility.

Sentence-specific analysis: Inspect the technical word in the original Telugu input, consonant clusters and vowel length. The rule-based Unicode frontend deterministically maps characters to phones, but that mapping alone cannot establish natural technical-word pronunciation. The predicted F0 declines from the opening to the ending token window.

Duration: median 6.5 frames; longest 16 frames (0.186 s) at token 37. **F0:** peak 321.95 Hz at token 4; first/last five-token means 276.6/159.9 Hz. **Energy:** maximum 140.61 at token 1, minimum 8.16 at token 16. These positions identify timing/emphasis candidates and weak-phone/pause regions for listening, rather than proving lexical stress. Token means are unweighted; phone identity and F0 interpolation affect the curves.

Pronunciation / overall quality: listening review pending. Inspect the target words, omissions/repetitions, naturalness and phrase timing in the exact linked WAV; use the interactive companion to record notes. Predicted F0 is phone-level and interpolated, not a measured generated-waveform F0 track.

Original labeled plots: Duration | Pitch | Energy | All sample links

15 / 5 / PER-SENTENCE EVALUATION

T2a: Grip/hold sense of pattu

Telugu | acoustic step 13,000 | speaker 0 | Play / download WAV | Raw text and arrays

Raw assignment input:

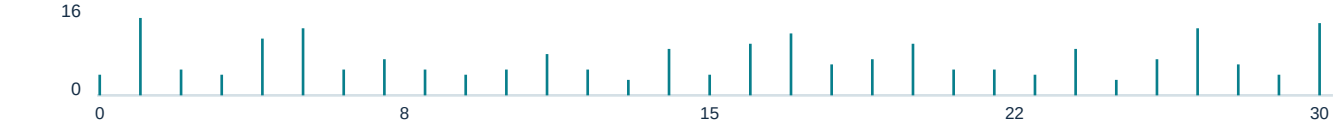
తాడుపై అతని పట్టు చాలా బలంగా ఉంది.

Saved normalized model input:

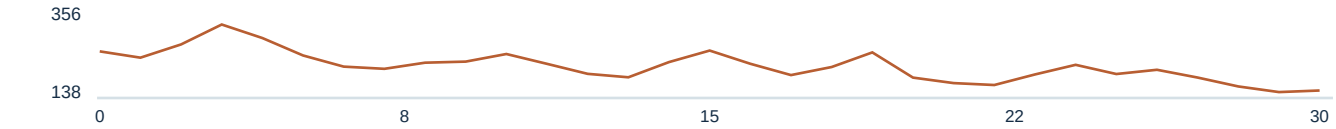
తాడుపై అతని పట్టు చాలా బలంగా ఉంది.

Measured waveform / prediction	Value
Tokens / frames / WAV duration	31 / 222 / 2.577 s
F0 mean [range]; energy mean [range]	221.29 [153.51, 329.22] Hz; 59.73 [4.07, 130.15]
Duration median / max; zero phones; pause frames	6 / 15; 0; 0

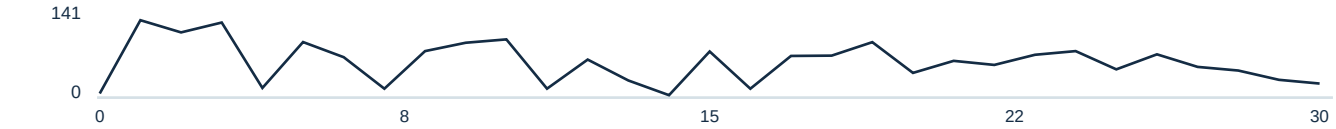
Duration / mel frames

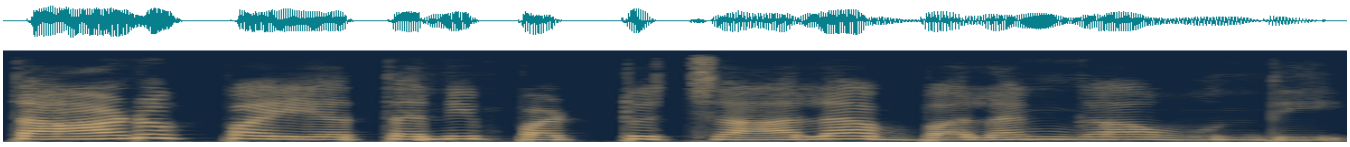


Predicted phone F0 / Hz



Predicted phone energy / STFT units





Waveform + 0-8 kHz spectrogram; Hann 1,024 / hop 128; colors are dB relative to utterance peak.

Measured audio: RMS -16.6 dBFS; peak -3.2 dBFS; 0 full-scale samples. Nonzero energy and no full-scale saturation do not establish intelligibility.

Sentence-specific analysis: The target word has the grip/hold sense, while T2b uses the silk sense. Both require the same orthographic consonant cluster/geminate handling. Compare timing and phrase contour rather than assuming the acoustic model encodes semantic understanding.

Duration: median 6 frames; longest 15 frames (0.174 s) at token 1. **F0:** peak 329.22 Hz at token 3; first/last five-token means 280.6/176.4 Hz. **Energy:** maximum 130.15 at token 1, minimum 4.07 at token 14. These positions identify timing/emphasis candidates and weak-phone/pause regions for listening, rather than proving lexical stress. Token means are unweighted; phone identity and F0 interpolation affect the curves.

Pronunciation / overall quality: listening review pending. Inspect the target words, omissions/repetitions, naturalness and phrase timing in the exact linked WAV; use the interactive companion to record notes. Predicted F0 is phone-level and interpolated, not a measured generated-waveform F0 track.

Original labeled plots: Duration | Pitch | Energy | All sample links

16 / 5 / PER-SENTENCE EVALUATION

T2b: Silk sense of pattu

Telugu | acoustic step 13,000 | speaker 0 | Play / download WAV | Raw text and arrays

Raw assignment input:

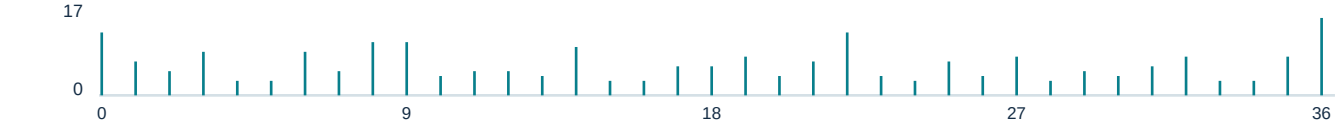
ఆమె పట్టు చీరను ధరించి వేడుకకు వచ్చింది.

Saved normalized model input:

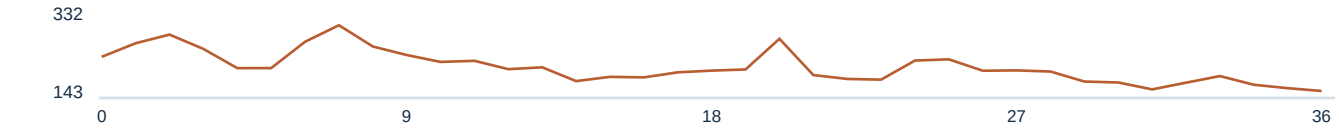
ఆమె పట్టు చీరను ధరించి వేడుకకు వచ్చింది.

Measured waveform / prediction	Value
Tokens / frames / WAV duration	37 / 236 / 2.740 s
F0 mean [range]; energy mean [range]	213.02 [158.50, 307.49] Hz; 55.04 [5.65, 133.13]
Duration median / max; zero phones; pause frames	5 / 16; 0; 0

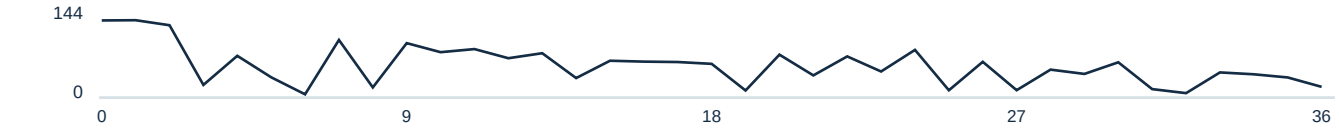
Duration / mel frames

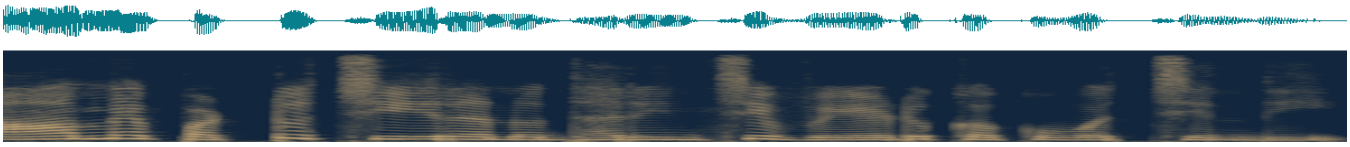


Predicted phone F0 / Hz



Predicted phone energy / STFT units





Waveform + 0-8 kHz spectrogram; Hann 1,024 / hop 128; colors are dB relative to utterance peak.

Measured audio: RMS -17.1 dBFS; peak -3.7 dBFS; 0 full-scale samples. Nonzero energy and no full-scale saturation do not establish intelligibility.

Sentence-specific analysis: This sentence provides the silk context for the same written target word used in T2a. The saved sentence has no explicit sp tokens; phrase timing is represented by phone durations. Correct geminate realization and natural contextual prosody require Telugu-proficient listening.

Duration: median 5 frames; longest 16 frames (0.186 s) at token 36. **F0:** peak 307.49 Hz at token 7; first/last five-token means 250.6/173.1 Hz. **Energy:** maximum 133.13 at token 1, minimum 5.65 at token 6. These positions identify timing/emphasis candidates and weak-phone/pause regions for listening, rather than proving lexical stress. Token means are unweighted; phone identity and F0 interpolation affect the curves.

Pronunciation / overall quality: listening review pending. Inspect the target words, omissions/repetitions, naturalness and phrase timing in the exact linked WAV; use the interactive companion to record notes. Predicted F0 is phone-level and interpolated, not a measured generated-waveform F0 track.

Original labeled plots: Duration | Pitch | Energy | All sample links

17 / 5 / PER-SENTENCE EVALUATION

T3: Broken Telugu word

Telugu | acoustic step 13,000 | speaker 0 | Play / download WAV | Raw text and arrays

Raw assignment input:

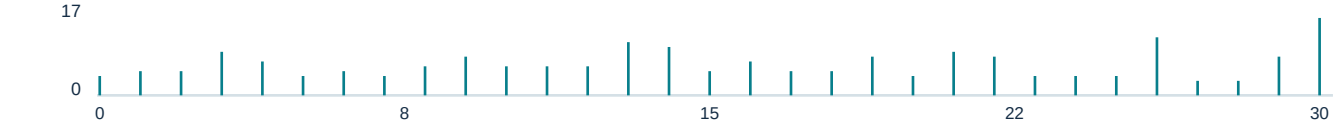
పులి అడ విలో నిశ్శబ్దంగా నడిచింది.

Saved normalized model input:

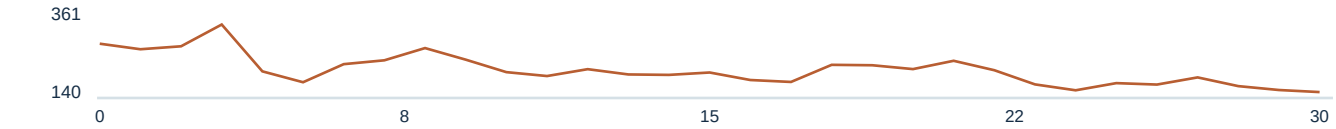
పులి అడ విలో నిశ్శబ్దంగా నడిచింది.

Measured waveform / prediction	Value
Tokens / frames / WAV duration	31 / 201 / 2.334 s
F0 mean [range]; energy mean [range]	214.33 [155.94, 334.33] Hz; 63.36 [5.64, 115.50]
Duration median / max; zero phones; pause frames	6 / 16; 0; 0

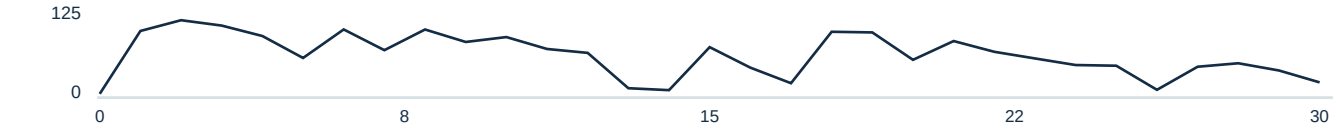
Duration / mel frames

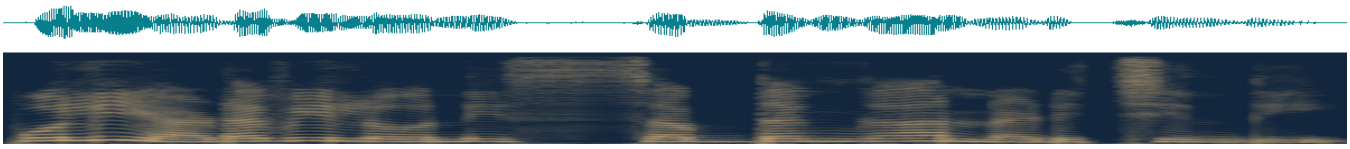


Predicted phone F0 / Hz



Predicted phone energy / STFT units





Waveform + 0-8 kHz spectrogram; Hann 1,024 / hop 128; colors are dB relative to utterance peak.

Measured audio: RMS -16.5 dBFS; peak -3.1 dBFS; 0 full-scale samples. Nonzero energy and no full-scale saturation do not establish intelligibility.

Sentence-specific analysis: The raw input retains the double space in the assignment broken word. Normalization collapses repeated spaces without joining the pieces. No explicit sp token is present, so source spacing must not be interpreted as a measured pause. Check whether the artificial boundary changes word intelligibility.

Duration: median 6 frames; longest 16 frames (0.186 s) at token 30. **F0:** peak 334.33 Hz at token 3; first/last five-token means 274.9/171.9 Hz. **Energy:** maximum 115.50 at token 2, minimum 5.64 at token 0. These positions identify timing/emphasis candidates and weak-phone/pause regions for listening, rather than proving lexical stress. Token means are unweighted; phone identity and F0 interpolation affect the curves.

Pronunciation / overall quality: listening review pending. Inspect the target words, omissions/repetitions, naturalness and phrase timing in the exact linked WAV; use the interactive companion to record notes. Predicted F0 is phone-level and interpolated, not a measured generated-waveform F0 track.

Original labeled plots: Duration | Pitch | Energy | All sample links

18 / 5 / PER-SENTENCE EVALUATION

T4: Year, ordinal, time and clause boundary

Telugu | acoustic step 13,000 | speaker 0 | Play / download WAV | Raw text and arrays

Raw assignment input:

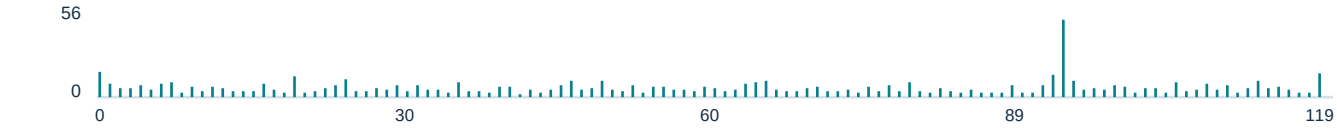
ఈ సమావేశం 2026 నవంబర్ 12వ తేదీ ఉదయం 10:45 గంటలకు ప్రారంభమవుతుంది, సమయానికి సిద్ధంగా ఉంచండి.

Saved normalized model input:

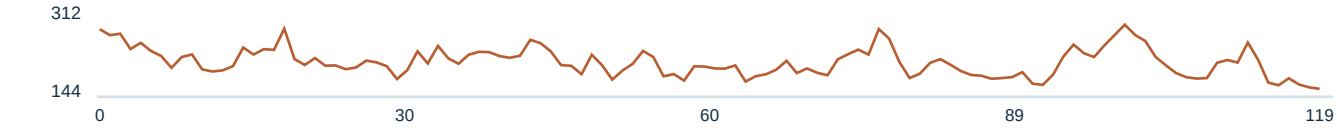
ఈ సమావేశం రెండు వేల ఇరవై ఆరు నవంబర్ పన్నెండవ తేదీ ఉదయం పది నలభై ఐదు గంటలకు ప్రారంభమవుతుంది, సమయానికి సిద్ధంగా ఉంచండి.

Measured waveform / prediction	Value
Tokens / frames / WAV duration	120 / 775 / 8.998 s
F0 mean [range]; energy mean [range]	213.89 [160.12, 289.32] Hz; 56.93 [1.38, 114.45]
Duration median / max; zero phones; pause frames	5 / 52; 0; 52

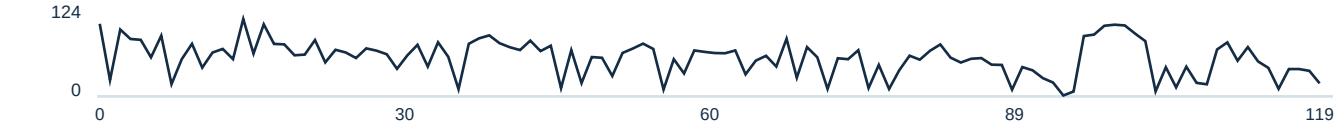
Duration / mel frames

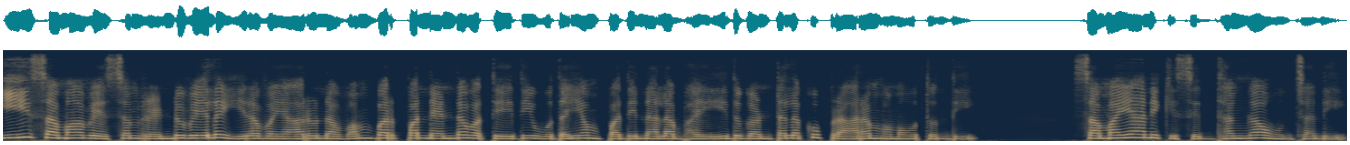


Predicted phone F0 / Hz



Predicted phone energy / STFT units





Waveform + 0-8 kHz spectrogram; Hann 1,024 / hop 128; colors are dB relative to utterance peak.

Measured audio: RMS -17.1 dBFS; peak -3.3 dBFS; 0 full-scale samples. Nonzero energy and no full-scale saturation do not establish intelligibility.

Sentence-specific analysis: This is the longest Telugu sample: 120 tokens and 775 mel frames. The normalized text expands the year, ordinal date and time into Telugu words. A single sp token at index 94 expands for 52 frames (0.604 s), marking the comma/clause break; assess whether the numeric grammar and pause sound appropriate.

Duration: median 5 frames; longest 52 frames (0.604 s) at token 94. **F0:** peak 289.32 Hz at token 100; first/last five-token means 262.5/168.1 Hz. **Energy:** maximum 114.45 at token 14, minimum 1.38 at token 94. These positions identify timing/emphasis candidates and weak-phone/pause regions for listening, rather than proving lexical stress. Token means are unweighted; phone identity and F0 interpolation affect the curves.

Pronunciation / overall quality: listening review pending. Inspect the target words, omissions/repetitions, naturalness and phrase timing in the exact linked WAV; use the interactive companion to record notes. Predicted F0 is phone-level and interpolated, not a measured generated-waveform F0 track.

Original labeled plots: Duration | Pitch | Energy | All sample links

19 / 6 / CROSS-LANGUAGE EVALUATION

Comparison, limitations and improvements

Dimension	English	Telugu
Source / conditioning	Accent-diverse ASR recordings; 115 demographic training profiles.	Studio TTS recordings; two source voice labels.
Submitted synthesis	8k checkpoint / profile 14.	13k checkpoint / female voice 0.
Frontend evidence	Same noun-stress conduct in E2a/E2b; broken rhyth m; numeric hyphen inserts sp.	Rule-based Unicode phones; broken word; date/time/ordinal expansion.
Predicted variance	E1 has one zero-duration phone; E4 has 76 pause frames in two sp tokens.	No zero-duration phones in five saved arrays; T4 has one 52-frame sp token.
Generalization	Validation minimum 2.702820, then plateau despite lower training loss.	Validation minimum 1.965954, then plateau despite lower training loss.
Quality conclusion	Pronunciation/intelligibility/naturalness pending listening.	Same; review by a Telugu-proficient listener is needed.

What the evidence supports

All ten submitted WAVs are 22,050 Hz mono 16-bit PCM, nonempty and consistent with predicted frame totals. All three predictor arrays align with the saved phone list, and their duration sums match mel frames. The report includes measured RMS/peak levels and checks full-scale saturation. These checks establish structural/audio-signal properties; they do not prove correct words, natural speech or lexical stress.

Known failures and remaining evidence

Earlier alignment code omitted empty interior pauses while trimming retained their audio, shifting later phone/frame correspondence. Current source preserves gaps, rejects invalid spans and unknown-speech spn, and uses selected-training-only statistics. The earlier report also records 82 historical Telugu target/frontend mismatches (73 train / 9 validation). Those targets and trained weights need separate review; code corrections alone do not repair historical artifacts.

Recovered manifests have zero utterance-ID overlap and zero overlap among known waveform hashes. Final waveform hashes are missing for 382 retained English reserve clips and 339 Telugu clips. Full manifests and acoustic runtime helpers are omitted from this repository snapshot. Public Drive downloads, original hardware/environment, and completed per-sample listening records are not established by this report build.

Evidence-led improvement sequence

1. Restore missing runtime files and verify all model/metadata/vocoder checksums. 2. Repair alignment/target disagreements and complete final split-hash evidence. 3. Review real-mel vocoder reconstructions and compare reference-conditioned versus free-running held-out audio. 4. Correct context-sensitive English stress and numeric punctuation; validate Telugu allophones, clusters and numeric grammar. 5. Use held-out speech plus losses for checkpoint selection, then consider richer pitch modeling or vocoder adaptation. More steps alone are not justified by the current plateaus.

Comparison limit: corpus recording conditions, speaker characteristics and feature normalization differ. Lower Telugu losses or higher F0 values cannot establish a causal language advantage or higher naturalness. No MOS, WER, speaker-similarity or listening scores are invented.

20 / 7 / REPRODUCTION & REFERENCES

Reproducibility, commands and evaluator access

Colab pipeline | Main Drive | English config / Telugu config | Dependencies

Current execution blockers: scripts import run_support from project/tools/, which is absent. Colab also references omitted restore_artifacts.py, colab_pipeline.py, probe_audio.py, model_artifacts.json and Results/test_sentences.json, and expects root-level language paths instead of project/. Restore matching helpers and correct notebook paths before running. The commands below describe a restored environment; this report build does not claim GPU inference or retraining.

```
git clone https://github.com/SandeshAwate6055/FastSpeech2-English-Telugu.git
cd FastSpeech2-English-Telugu
python -m pip install -r project/requirements.txt
# Restore shared helpers, matching metadata, acoustic and vocoder weights.
cd project/English
python prepare_data.py --train_hours 5.0 --val_hours 1.0 --seed 42
# Obtain compatible MFA phone TextGrids before preprocessing.
python preprocess.py --config config.yaml
# Validate features/provenance and held-out speech before full training.
python train.py --config config.yaml --total_step 60000
python inference.py --config config.yaml \
  --ckpt_file checkpoints/8000.pth.tar --speaker_id 14 \
  --result_path ../Results/reproduced/English
# Repeat independent preparation, alignment and training from ../Telugu.
cd ../Telugu
python inference.py --config config.yaml \
  --ckpt_file checkpoints/13000.pth.tar --speaker_id 0 \
  --result_path ../Results/reproduced/Telugu
```

Inference omits --text to synthesize all five built-in assignment sentences and saves WAVs, three plots per sample and prediction JSON. Use fresh result paths to preserve supplied artifacts. MFA is a separate prerequisite; prepare_data.py alone does not produce TextGrids. New aligner fitting must exclude validation audio. Svarah may require Hugging Face approval/HF_TOKEN. Install NLTK tagger resources and cmudict as described in the README.

Model restoration: English 8k download / Telugu 13k download. Keep original speakers.json, stats.json, symbol inventory/lexicon and universal HiFi-GAN weights coupled to each checkpoint. English hashes/metadata / Telugu hashes/metadata. The companion includes exact checkpoint hashes. External downloads were not revalidated here.

Software: acoustic requirements use lower bounds, not a version lock (PyTorch/torchaudio, NumPy/SciPy/librosa, PyWorld, tgt, g2p_en, datasets, PyYAML, etc.). Record tested Python, CUDA/PyTorch, MFA and dataset revisions for reproduction. Report-only build uses reportlab, Pillow and NumPy; rendering checks use pypdf/pypdfium2. Report build and HTML usage

References: [1] Assignment Sections 1-4. [2] FastSpeech 2 paper, supplied v8, Section 2.3 and Appendix A-B. [3] ming024 reference. [4] Svarah and IndicTTS Telugu. [5] Measured report data; current source, CSVs, WAVs and prediction JSON linked throughout.